Steady Spoon - Electronic design

Excerpt of the Final Report (Group 10, Biodesign A.Y. 2024-2025, UCBM, v4.0, 10/01/2025) and of the attachments on the electronic design.

Contents: Sec. 2.3-2.4 of the report (electronic design); PCB Design, Charging Circuit, Voltage Regulators, Batteries, Microcontroller and Sensing attachments.

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Mechanics, motors, data analysis, firmware and the other attachments were developed by the team and are not included here. The full report is a group work.

and lower sections, not only facilitates the organization of all electrical components within the handle but also ensures ease of access and functionality during maintenance. As previously mentioned, the motor located distal from the handle and proximal to the tip of the spoon is safeguarded by a cover, which is magnetically coupled to the handle using neodymium magnets. The end of the casing is constructed from a different material to minimize mechanical resistance to the vertical movement of the tip. The selection of materials will be further discussed in Section 2.3.

- **Heat dissipation**: To promote effective heat dissipation from the batteries, a gap is incorporated between the PCB and the battery housing, reducing the risk of overheating.
- **Efficient connections between components**: The spoon tip is attached to the rest of the spoon through a joint with a specific geometry, ensuring proper alignment and positioning of the tip.
- **Informative**: The presence of two LEDs on the upper side of the handle allows SteadySpoon to inform the user about the battery status and charging progress.

A more comprehensive description of the mechanical design and its components is given in the *CAD Attachment*.

2.3 Materials and Manufacturing

The design of the spoon utilizes two innovative materials: polypropylene (PP) and Dragon Skin 10 Fast, each chosen for its unique properties to meet the specific requirements of the project.

Polypropylene (PP), a biocompatible and hypoallergenic thermoplastic polymer, is employed to manufacture the entire spoon except for the tip-close extremity of the cover. PP properties allow it to meet the requirements of lightweight, resistance to fatigue and abrasion, suitable for dishwasher cleaning, fire-proof, and electrical insulation. For manufacturing the PP components, the Multi Jet Fusion (MJF) 3D printing technology was chosen. This sintering process uses fusion and detailing agents to produce high-precision components with smooth surfaces and complex geometries. For portion of the tip-close extremity of the handle, Dragon Skin 10 Fast, a highly flexible and durable liquid silicone, was selected. Known for its softness and skin-safe properties, it is ideal for ensuring the spoon's internal electrical components insulation without hindering the tip's movement. Production is carried out using the casting and vulcanization technique, which allows for the rapid transformation of the liquid silicone into a solid state. For further informations see *Materials and Manifacturing Attachement*.

2.4 Electronic Design

The sensing and control module and the power source have been integrated on a PCB designed on *Autodesk Eagle*. This approach leads to an optimal dimensioned

board with size 40 x 85 mm, ideal for the CAD fitting. The components arranged on the PCB are:

- **Battery Pins:** To power the PCB;
- **Recharging Circuit:** Responsible for managing the battery recharging process;
- **5 V and 6 V Regulator:** Ensures a stable 5 V supply for the system;
- **9 V Charging Jack:** Connects to the charging circuit;
- **USB to Serial UART Interface (FT232RL):** Used for loading the firmware and supplying 3.3 V regulation;
- **Microcontroller:** `ATMega328-MUR`, which acts as the main controller of the platform;
- **USB-B 2.0 Connector:** For interfacing with external devices;
- **MPU-6050 Sensor:** Integrated to handle motion sensing;
- **Servo Pins:** Six pins for two servos; each servo has one pin for the power, one for the ground and one for PWM signals.

The board was designed by maintaining Arduino Nano components which satisfy the needs and adding specific ones to complete the task. Through this way of thinking space and power have been optimised. For an in depth description of the rationale behind the positioning of the components, please refer to the *PCB Attachment*.

2.4.1 Sensing Module

The sensing module in the design of the spoon has the aim to detect both rotational and linear tremor movements generated by the patients during their meal. For this purpose the IMU MPU-6050 have been selected. This integrated 6-axis Motion-Tracking device combines a 3-axis gyroscope and a 3-axis accelerometer and its high accuracy, sensitivity and fair full-scale range ensure precise detection of even the slightest tremors. The sensor has an on-chip Digital Motion Processor that enables real-time signal adjustments with minimal latency. Nonetheless MPU-6050 strenght points remain its low power consumption and compact size that make it suitable for integration into a lightweight, portable spoon design and its robustness that ensures durability and reliability essential for daily use by patients.Furthermore, its flexible I2C interface allows, through two pull-up resistors, an easy integrating component for the device. For more informations consult *Sensing Attachment*.

2.4.2 Actuation Module

The actuation module was designed to compensate the pitch and roll angles generated by tremors. The servos have been selected based on torque, angular velocity, power output, response time, size, and mass, considered essential parameters for

ensuring the device's effectiveness, as well as cost considerations to make the device as economical as possible. The motor design was developed under the assumption of a trapezoidal velocity profile, constructed based on prior data analysis, from which the required angular velocity was derived. Through the modeling of geometries and the calculation of moments of inertia, the necessary torque values were established. Two Emax servo models were selected: the ES3059MD and the ES9250MD. Both of them resulted to supply greater torques and velocities than need even at full load. Their rapid response, which enables the selected trapezoidal profile, and adequate electrical power make them capable of compensating the tremor. All this considering an easy control in position, good accuracy and restrained dimensions and price. For a complete dissertation consult *Motor Attachment*.

2.4.3 Control Module

The microcontroller selected for the application is the ATMega328-MUR. It communicates with the sensor with a I2C protocol and generates PWM signals to the motors. More informations are available in *Microcontroller Attachment*.

2.4.4 Power Source

The battery is the core power source for the spoon and must provide the necessary voltage and current for all the integrated devices: since the components have different voltage and current requirements, it was necessary to include voltage regulators (See *Regulators Attachment*). Two rechargeable Tenergy 18650 batteries were chosen, each with a capacity of 2200 mAh and a nominal voltage of 3.7 V, connected in series to achieve a total voltage of 7.4 V. A charging jack, compatible with 9 V chargers, can be connected to the power grid and to a charging circuit permitting the charge of the battery. The battery follows a CC/CV (constant current/constant voltage) charging protocol with a rapid charge current of 1 CA (2200 mA) and a maximum voltage of 8.4 V, with an estimated charging time of about 60 minutes (1 hour). The battery, considering the maximum current absorbed by the PCB, is fully discharged in 1 hour and 37 minutes. Safety tests under extreme conditions confirm that the battery is safe, with no risk of fire or explosion. The complete material is collected in *Regulators Attachment* , *Batteries Attachment* and *Charging circuit Attachment*.

2.5 Firmware Description

Figure 4 provides a visual representation of the logic flow chart that defines the structure of the firmware.

The developed firmware integrates several core functionalities, operating in a structured and efficient manner to control the system. The IMU sensor readings serve as the sensing aspect of the system, providing critical data for detecting tremors. The computation of roll and pitch angles constitutes the system's processing phase, transforming raw sensor data into actionable information. Finally, the control of the servomotors represents the acting phase, where the computed angles are used to mitigate tremors and stabilize the spoon. Key components of the

PCB Design Attachmnet

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The PCB was designed as shown in the Figure 1.

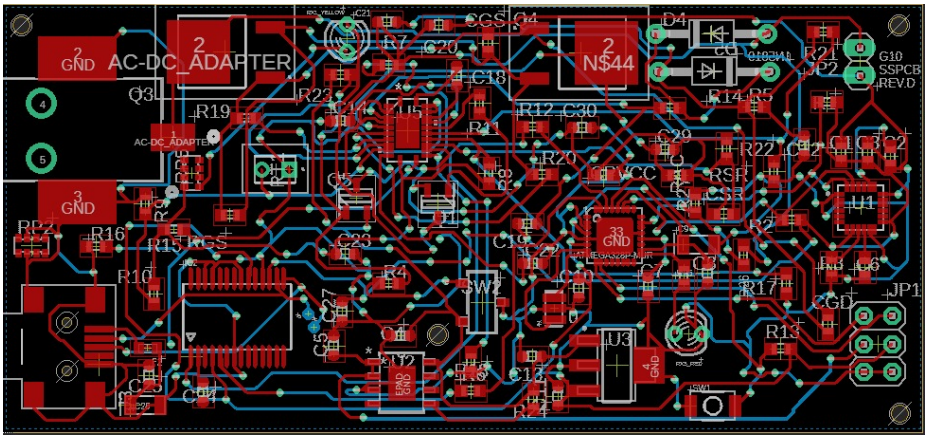


Figure 1: Component layout on the PCB

At the bottom center, a slide switch allows the device to be turned on and off. On the short left side, the USB input ports are located at the bottom and the jack input port at the top. A yellow LED, positioned at the top left, indicates the device's charging status (for more details please refer to the *Batteries Attachment*), while a red LED, located at the bottom right, signals a low battery, i.e., when the battery level drops below 7V. The PCB includes four holes used to secure the PCB to the case.

In the Figure 2, the schematic of the entire circuit can be seen. Below is the discussion of the various modules

Battery: PINHD-1X2

The battery is connected to the PCB via two dedicated pins: one is connected to the ground, while the other is connected to the positive power line (V_{batt}), which carries the voltage supplied by the battery to the circuit. This connection is made through pinhead connectors. Further details can be found in *Batteries Attachment*

Charging Module: BQ24172RGYR

The charging module has been subject of its own attachement where the configuration of the major pins and design of the electrical components depending on the desired specifics have been discussed.

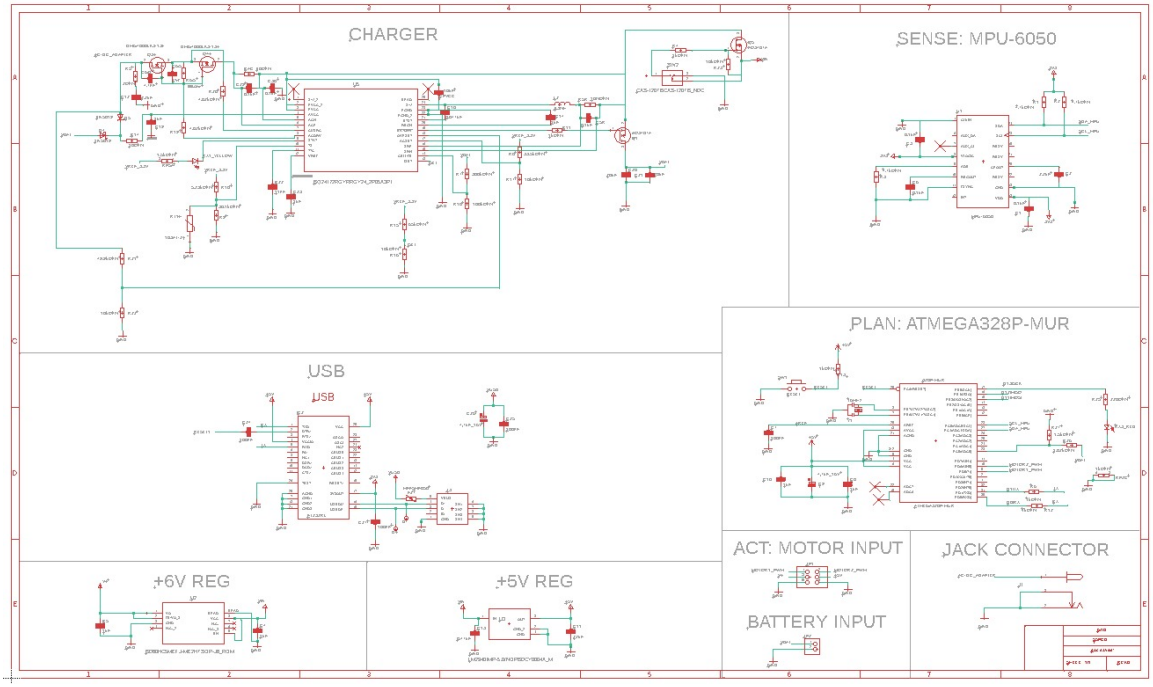


Figure 2: Schematic of the circuit

Hence for more clues please consult *Charging Module Attachment*. In the Figure 3, the schematic of the charging module is shown. The charging module is connected to the outside via an AC/DC adapter with a jack cable.

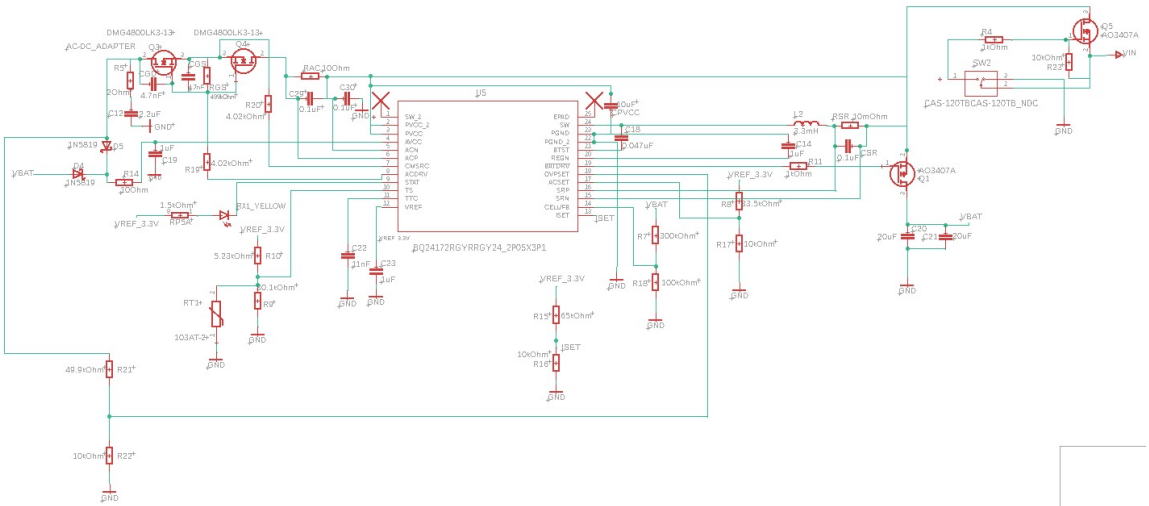


Figure 3: Schematic of the charging module

Charging Jack

A charging jack, compatible with 9 V chargers, can be connected to the power grid and to a charging circuit permtting the charge of the battery The charging jack has heating issues, so there are no traces under it. With no traces beneath the IC, heat is dissipated more effectively, reducing the risk of overheating, while minimizing electrical interference that could negatively affect the component's performance.

Microcontroller: ATMEGA328P-32MUR

The schematic of the microcontroller, derived from the component's datasheet [3], is shown in the Figure 4. Note that not all available pins have been used: the unused pins have been left floating.

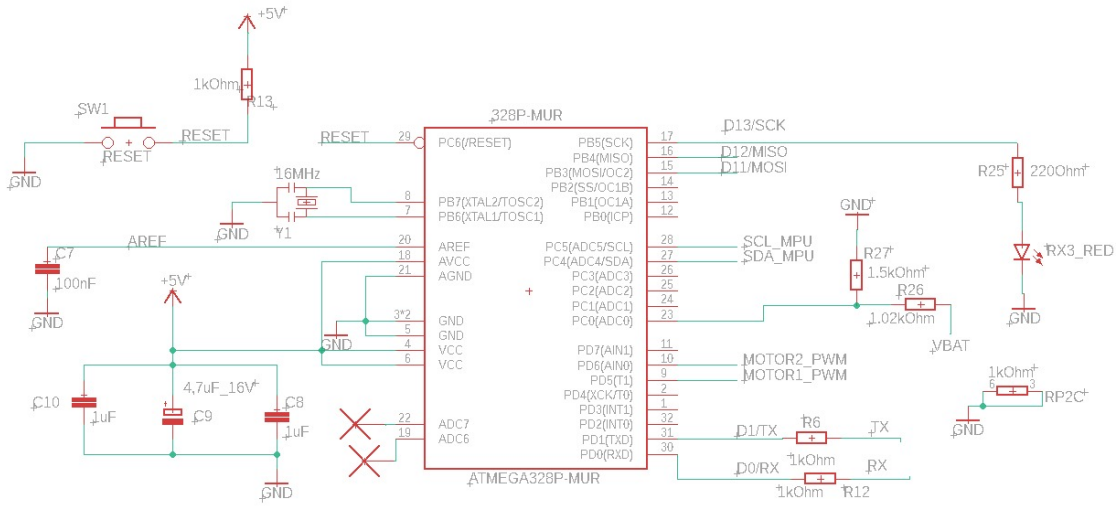


Figure 4: Schematic of the microcontroller

To ensure proper power supply and reduce electrical noise, the bypass capacitors (C7, C8, C9, C10) are placed on the PCB as close as possible to the ATMega328P-32MUR microcontroller. The external oscillator, which provides a stable and precise 16MHz clock signal to synchronize the microcontroller's operations, is also positioned near the microcontroller to minimize interference and ensure better signal quality. The resistors R6 (pull-up) and R12 (pull-down) ensure reliable communication between the microcontroller and the USB serial, properly managing the transmission and receiving lines. Communication with the IMU sensor is achieved through the SDA and SCL lines, enabling the I2C protocol for data exchange. Two dedicated pins are used for the generation of PWM signals to independently control the motors. Finally, the RESET pin, connected to a reset button, allows for the microcontroller to be restarted when needed.

Sensor: MPU-6050

In the Figure 5, the Typical Operating Circuit of the MPU6050 sensor [2] can be observed

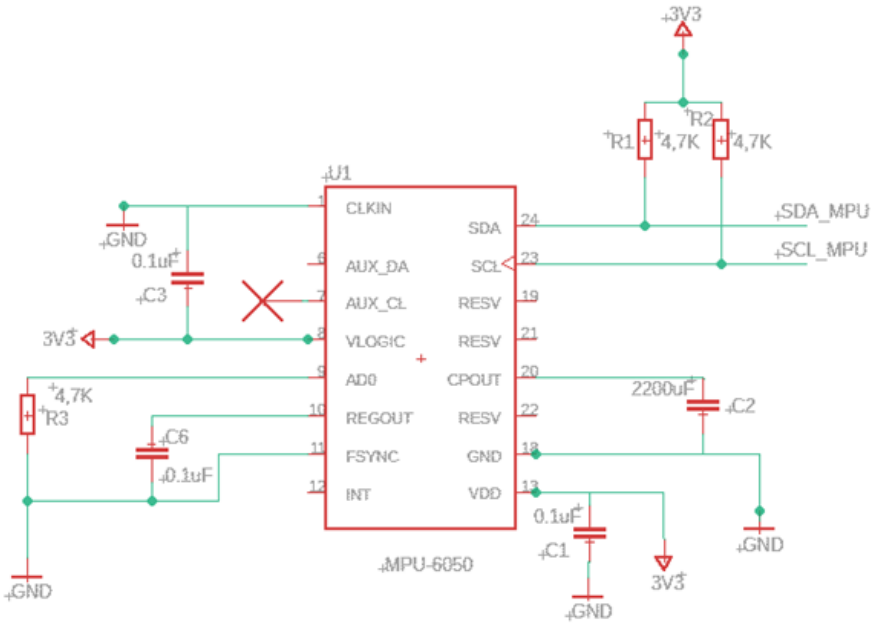


Figure 5: Typical Operating Circuit of the MPU6050 sensor

The project involves careful placement of this component on the PCB to optimize performance and reduce interference. The sensor is positioned as far as possible from the center of the board, along the longest axis, to avoid noise and interference. Additionally, the sensor is centered and aligned with the index finger, which has been identified as the reference point for data collection. The bypass capacitors (C1, C2, C3, and C6) are placed as close as possible to the sensitive components.

Regulators: BD60HC5MEFJ-ME2 (6V) and LM2940IMP-5.0/NOPB (5V)

In the Figure 6 and Figure 7, the Typical Operating Circuit of the regulators ([4], [1]) is shown. In the PCB design, no traces are placed under the regulators to improve heat dissipation and prevent electrical interference. In this case as well, the bypass capacitors are placed as close as possible to the components.

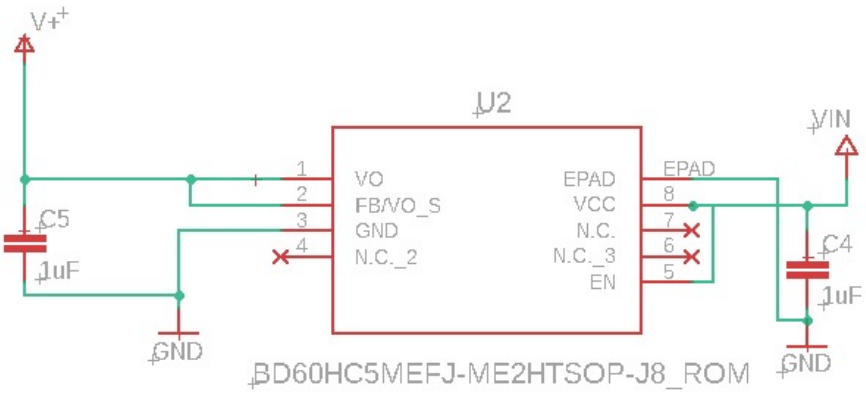


Figure 6: Typical Operating Circuit of BD60HC5MEFJ-ME2

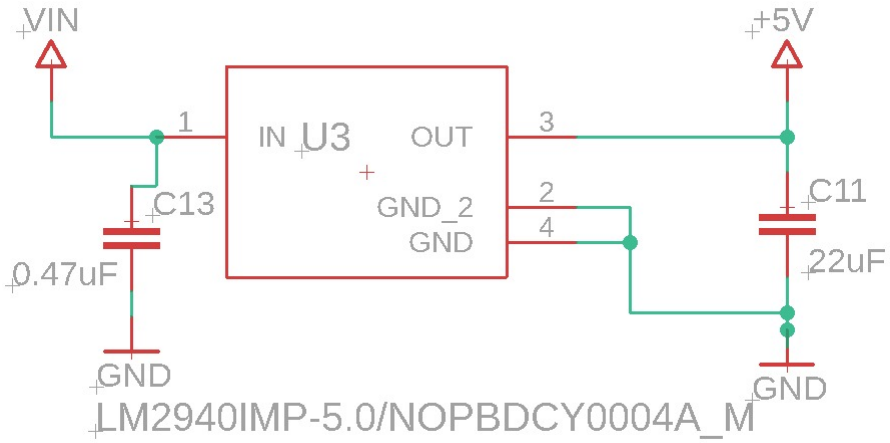


Figure 7: Typical Operating Circuit of LM2940IMP-5.0/NOPB

USB Module: FT232RL

The USB module consists of an FT232R chip and a female USB Type B 2.0 connector. The FT232R serves as a USB-to-serial UART converter and includes a built-in 3.3 V low-dropout (LDO) voltage regulator. It connects to the Atmega328 microcontroller using its RX and TX pins and is also linked to the reset button for communication or programming. The module operates on a 5 V power supply, delivered via the LM1117 voltage regulator to its VCC pin, and typically consumes 15 mA of current during normal operation. According to the datasheet, the 3.3 V output from the integrated LDO regulator needs to be decoupled to ground using a 100 nF capacitor (C27). This output can also be used to supply external circuits requiring a +3.3 V nominal voltage, with a maximum current of 50 mA. A capacitor C24 is connected to the reset button to reduce the effects of switch bouncing. Both C24 and C27 are positioned close to the FT232R chip. Additionally, two more bypass capacitors

(C26 and C25) are placed near the USB connector to enhance stability.

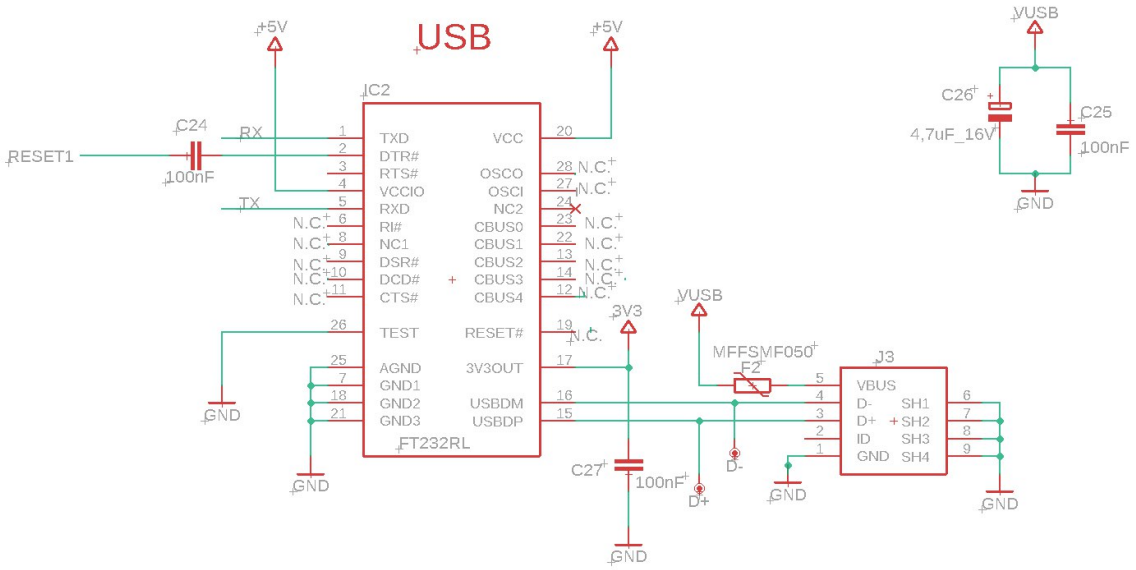


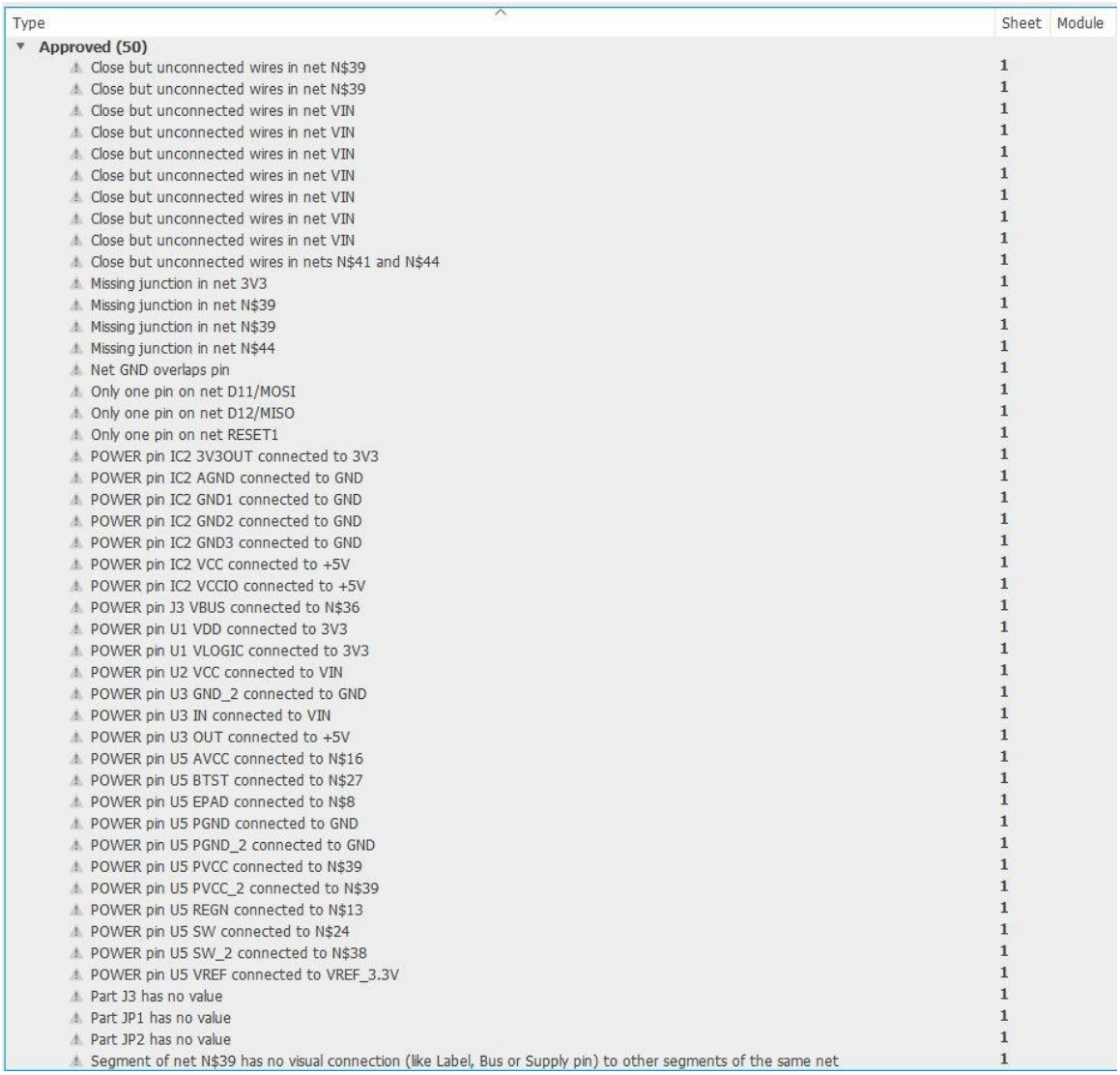
Figure 8: Schematic of the USB Module

Emax Servos

The ES3059MD and the ES9250MD servo are connected to the board by a six-pin configuration, arranged in a 2x3 header format. Each motor requires three pins: VCC, GND, and PWM.

Schematic and Board

On the schematic, there are 50 approved warnings, while on the board there are no errors.



Type	Sheet	Module
Approved (50)		
Close but unconnected wires in net N$39	1	
Close but unconnected wires in net N$39	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in net VIN	1	
Close but unconnected wires in nets N$41 and N$44	1	
Missing junction in net 3V3	1	
Missing junction in net N$39	1	
Missing junction in net N$39	1	
Missing junction in net N$44	1	
Net GND overlaps pin	1	
Only one pin on net D11/MOSI	1	
Only one pin on net D12/MISO	1	
Only one pin on net RESET1	1	
POWER pin IC2 3V3OUT connected to 3V3	1	
POWER pin IC2 AGND connected to GND	1	
POWER pin IC2 GND1 connected to GND	1	
POWER pin IC2 GND2 connected to GND	1	
POWER pin IC2 GND3 connected to GND	1	
POWER pin IC2 VCC connected to +5V	1	
POWER pin IC2 VCCIO connected to +5V	1	
POWER pin J3 VBUS connected to N$36	1	
POWER pin U1 VDD connected to 3V3	1	
POWER pin U1 VLOGIC connected to 3V3	1	
POWER pin U2 VCC connected to VIN	1	
POWER pin U3 GND_2 connected to GND	1	
POWER pin U3 IN connected to VIN	1	
POWER pin U3 OUT connected to +5V	1	
POWER pin U5 AVCC connected to N$16	1	
POWER pin U5 BTST connected to N$27	1	
POWER pin U5 EPAD connected to N$8	1	
POWER pin U5 PGND connected to GND	1	
POWER pin U5 PGND_2 connected to GND	1	
POWER pin U5 PVCC connected to N$39	1	
POWER pin U5 PVCC_2 connected to N$39	1	
POWER pin U5 REGN connected to N$13	1	
POWER pin U5 SW connected to N$24	1	
POWER pin U5 SW_2 connected to N$38	1	
POWER pin U5 VREF connected to VREF_3.3V	1	
Part J3 has no value	1	
Part JP1 has no value	1	
Part JP2 has no value	1	
Segment of net N$39 has no visual connection (like Label, Bus or Supply pin) to other segments of the same net	1	

Figure 9: Schematic Warnings

References

[1] TEXAS INSTRUMENTS. *LM2940 Low Dropout Regulator*. Datasheet. 2000. URL: https://www.ti.com/lit/ds/symlink/lm2940c.pdf.

[2] IvenSense. *IMU MPU6050*. Datasheet. 2013. URL: https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf.

[3] MICROCHIP. *megaAVR Datasheet*. Datasheet. 2020. URL: https://ww1.microchip.com/downloads/en/DeviceDoc/ATmega48A-PA-88A-PA-168A-PA-328-P-DS-DS40002061B.pdf.

[4] ROHM SEMICONDUCTOR. *Variable Output LDO Regulator BD60HC5MEFJ-M*. Datasheet. 2012. URL: https://xonstorage.blob.core.windows.net/pdf/rohm_bd70hc5vefjme2_apr22_xonlink.pdf.

Charging Circuit Attachment

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BQ24172

The selected charging module is the **bq24172**, an autonomous charger for lithium batteries (Li-ion and Li-Polymer) with power path management. This module utilizes a fixed-frequency synchronous PWM controller and provides high precision in regulating input current, charging current, and battery regulation voltage [1].

Charge Circuit

The device charges the battery in three phases: preconditioning, constant current, and constant voltage (Figure 1). The charging process terminates automatically when the current decreases to 10% of the programmed fast charge rate. For added safety, a programmable timer serves as a safeguard against overcharging. If the battery voltage drops below a set threshold, the bq24172 seamlessly restarts the charging cycle. Furthermore, to conserve energy, the device transitions into a low-power sleep mode when the input voltage falls below the battery voltage.

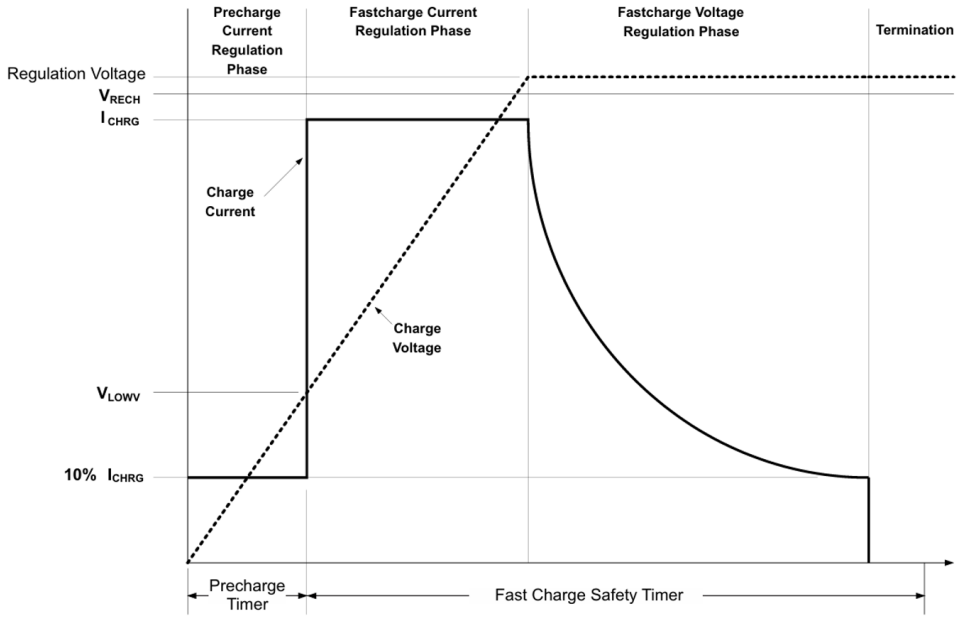


Figure 1: Typical Charging Profile.

Typical Charging Profile

Typical charging profile is now analysed and just its main charateristics and passive components associated are designed in the following subsections. The whole charging profile is valid between 0 and 45 °C.

Battery Voltage Regulation

In reference to Figure 1 Battery Voltage Regulation has been evaluated. To fully charge the series of two Tenergy Lithium Ion Cell, which have a maximum charge voltage of 4.2 V each, an electric tension of 8.4 V should be furnished. Associated resistors have been modeled in order to satisfy the requirment throughout the application of the following equation aviailable in the component datasheet.

$$V_{\mathrm{BAT}} = 2.1 \times \left(1 + \frac{R_7}{R_{18}}\right) = 8.4\,\mathrm{V}$$

R_7 and R_{18} ratio has to be 3:1 so their values are 300 kΩ and 100 kΩ respectively 2. Using these electrical resistances a tiny current is supposed to flow maintaining the precision of the system.

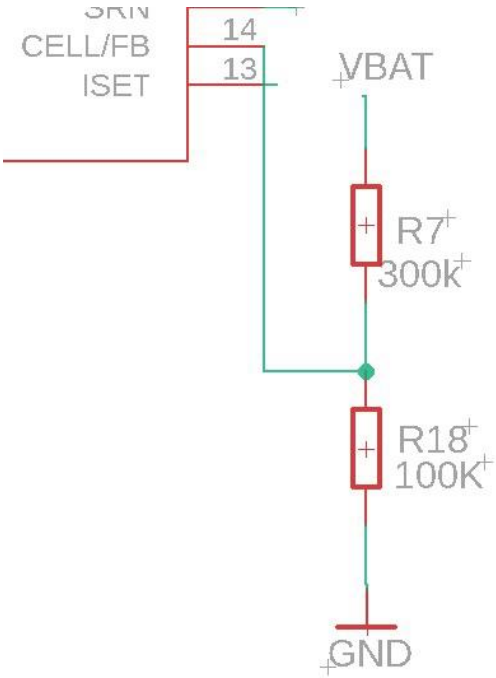


Figure 2: Resistors' value for Voltage Regulation.

Battery Current Regulation

The pin `ISET` sets the maximum charging current. The battery current is detected using the sensing resistor R_{SR} connected between the `SRP` and `SRN` pins. The equation for charge current is:

$$I_{\mathrm{CHARGE}} = \frac{V_{\mathrm{ISET}}}{20 \times R_{\mathrm{SR}}}$$

With a sensing resistor of $10\,\mathrm{m\Omega}$, the maximum output current is $4\,\mathrm{A}$. In order to have a fast charge of the device a maximum charging of 2.2 A (1 C of the battery) is needed and so an `ISET` voltage of 0.44 V is required. The voltage divider between V_{ISET} and V_{REF} gives the value of the involved resistences which are $R_{15} = 65\,\mathrm{k\Omega}$ and $R_{16} = 10\,\mathrm{k\Omega}$ (Figure 3).

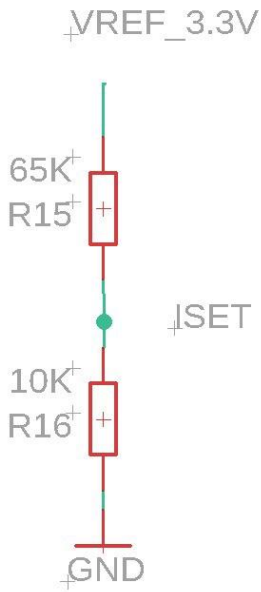


Figure 3: Resistors' value for Battery Current Regulation.

Input Current Regulation

The bq24172 incorporates Dynamic Power Management (DPM) to optimize the charging process and prevent overloading the power source. By using DPM, the input current regulator reduces the charging current when the summation of system power and charge power exceeds the maximum input power.

To regulate the input current, the `ACSET` pin defines the input current limit, calculated as:

$$I_{\mathrm{DPM}} = \frac{V_{\mathrm{ACSET}}}{20 \times R_{\mathrm{AC}}}$$

where I_{DPM} is:

$$I_{\mathrm{DPM}} = I_{\mathrm{CHARGE}} + I_{\mathrm{SYSTEM}} = 2.2\,\mathrm{A} + 1.35\,\mathrm{A} = 3.55\,\mathrm{A}$$

Where I_{SYSTEM} is determined in *Battery Attachment*. From I_{DPM}, we calculate $V_{\mathrm{ACSET}} = 0.76\,\mathrm{V}$. The value of the resistances result to be $R_8 = 33.5\,\mathrm{k\Omega}$ and $R_{17} = 10\,\mathrm{k\Omega}$ (FIgure 4). The input current is sensed across the R_{AC} resistor and its default value of R_{AC} is $10\,\mathrm{m\Omega}$.

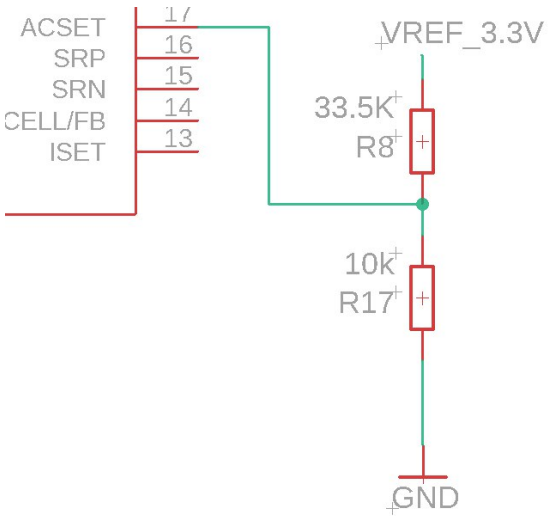


Figure 4: Resistors' value for DPM Regulation.

Charge Termination, Recharge, and Safety Timers

The `TTC` pin is used to protect the battery. To calculate the capacitor value C_{TTC}, the following formula is used:

$$\mathrm{t}_{\mathrm{TTC}} = C_{\mathrm{TTC}} \times K_{\mathrm{TTC}}$$

where t_{TTC} is the desired time of charge (60 minutes for the fast-charge) and $K_{\mathrm{TTC}} = 5.6\,\mathrm{min/nF}$ (as specified in the datasheet). Thus:

$$C_{\mathrm{TTC}} = \frac{t_{\mathrm{TTC}}}{K_{\mathrm{TTC}}} = 10.71\,\mathrm{nF}$$

which is rounded to the standard value of 11 nF.

System Power Selector

The System Power Selector in the bq24172 manages the system's power source by automatically switching between the adapter (when available) and the battery. By default, the system is powered by the battery during startup or when in sleep mode. When an adapter is connected and its voltage exceeds the voltage of the battery, the chip exits sleep mode, disconnects the battery and switches to the adapter as the power source. To ensure a safe and efficient management of the power sources N-channel MOSFETs (DMG4800LK3 MOSFET) are employed. The ACFET (Q3) controls the connection between the adapter and system and limits inrush current during adapter plug-in. While the RBFET provides reverse battery discharge protection. They are represented in Figure

5. A P-channel MOSFET (AO3407A MOSFET) is responsible for managing the connection between the battery and the system load (Q1 in Figure 6).

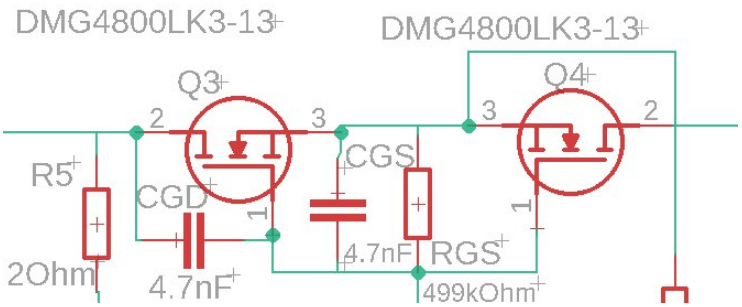


Figure 5: NMOSFETS.

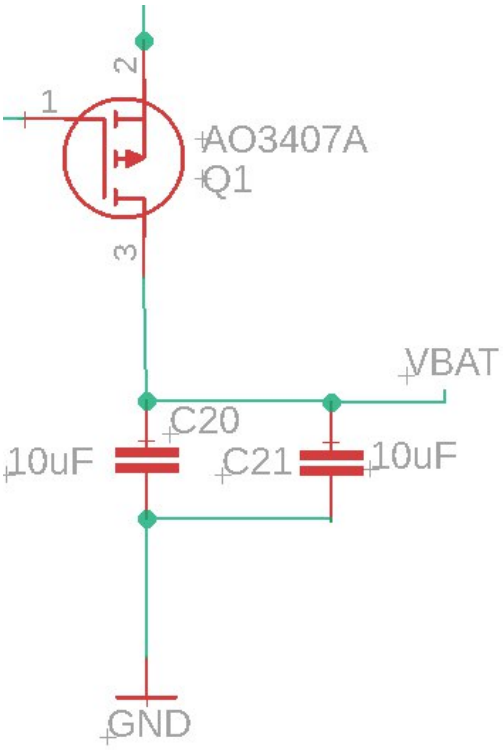


Figure 6: PMOSFET.

Charge Status Indicator

STAT pin drives a LED that warns the user about a charge in progress (light is still), charge suspended (light blinks) or charge complete (light is off) (Figure 7). In order to have a reduced electrical current a 1500 Ω resistance has been selected. The current flowing through the LED, considering a standard LED voltage of 2.0 V, is calculated as:

$$I_{\text{LED}} = \frac{V_{\text{REF}} - V_{\text{LED}}}{R_{10}} = \frac{(3.3\,\text{V} - 2.0\,\text{V})}{1500\,\Omega} \approx 0.87\,\text{mA}$$

The power consumed by the LED is:

$$P_{\text{LED}} = V_{\text{LED}} I_{\text{LED}} = 2 \cdot 0.87 \cdot 10^{-3}\,\text{W} = 1.75\,\text{mW}$$

I_{LED} is sufficient for visible illumination in dimly lit environments and ensures a low power consumption.

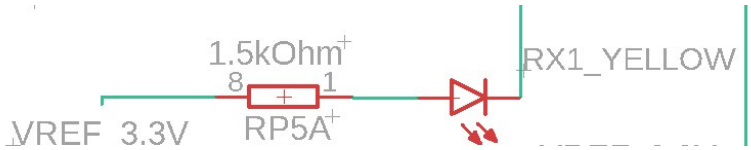


Figure 7: LED.

References

[1] Texas Instruments. *Charger With Integrated MOSFETs and Power Path Selector*. Datasheet. 2010. URL: https://www.ti.com/lit/ds/symlink/bq24170.pdf.

Voltage Regulators Attachment

GRUPPO 10

Voltage Regulators

The devices to power in the project are the two servos, the microcontroller and the sensor. Each of them has different supply requirements therefore to optimise voltage and current provided, the employment of voltage regulators was preferred.

Criteria of Selection

The pick of the voltage regulators has been very selective: only fixed exits voltage regulators with desired electric tension and appropriate current for all the devices connected have been evaluated. Other criteria must be satisfied for a proper selection of the regulator. The component has to be available in a SMD package, has to accept a 7.4 V input (which is the electrical tension of the battery), be avilable in small packages and have a low drop out voltage to guarantee a proper functioning of the component. Lastly for an appropriate electrical managment, the output current from the regulator has to be at least 20% greater than the sum of the input currents of the devices powered.

Electrical Supply Organization

All the following considerations were made after selecting a battery with adequate voltage and current capacity to meet the system's requirements, as shown in the *Batteries Attachment*.

The presence of different components demanding different voltages lead to the choice of 2 different regulators at 6.0 V and 5.0 V as shown in Figure 1 and written in Table 1.

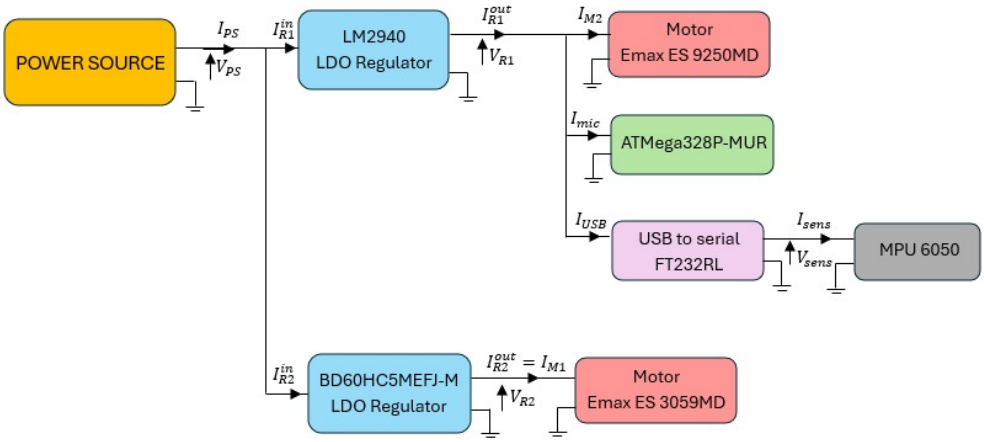


Figure 1: Components' supply Scheme.

Item	Value	Symbol
Power Source Output Current	2200 mA	I_{PS}
Power Source Output Voltage	7.4 V	V_{PS}
Fixed Voltage Output of LM2940	5.0 V	V_{R1}
Maximum Current exit of LM2940	1.0 V	I_{R1}^{out}
Maximum Input Current ES9250MD	330 mA	I_{M2}
Maximum Input Current ATMega	9 mA	I_{mic}
Maximum Input Current FT232RL	15 mA	I_{USB}
Maximum Input Current MPU-6050	3.9 mA	I_{sens}
Fixed Voltage Output of FT232RL	5.0 V	V_{sens}
Fixed Voltage Output of BD60HC5MEFJ	6.0 V	V_{R2}
Maximum Current exit of BD60HC5MEFJ	1.5 V	I_{R2}^{out}

Table 1: Voltage and current values.

As visible in Figure 1 every active component is powered through a voltage regulator except for the MPU-6050. Indeed the sensor exploits a specific pin of the USB UART interface Integrated Circuit Device FT232RL that can power external logics at 3.3 V up to 50 mA (3.9 mA are actually needed for the purpose). For this reason the 5V regulator supplies the USB Module and an additional voltage regulator at 3.3 V could be avoided. The other components powered at 5.0 V are the handle-close motor (M2) and the microcontroller. The common voltage of the elements pushed to unify their voltage regulators even though their electrical currents appear completely different (respectively 330 mA, 50 mA and 15 mA for the servo, the ATMega and the FT232RL). The assorbtion of the charge, consulting the components' datasheets, occurs autonomously ergo

no constraint limits the option.

Because of its 6V and 1A supply the motor closer to the spoon tip (M1) has been thought to have its own regulator. Each regulator will be briefly analised in its own subsection.

LM2940 Low Dropout Regulator

For the second servo, the microcontroller and the USB Module the decision ended up on the LM2940, a LDO voltage regulator by Texas Instrument [1]. In particular on the LM2940IMP-5.0/NOPB accessible in a SOT-223 type package (Figure 2). The following Table 2 is filled up with all utile the informations valid for a temperature range of -40 to +85 °C.



Figure 2: LM2940IMP-5.0/NOPB

Item	Value
Input Voltage	from 6 V to 26 V
Output Voltage	5.00 V
Output current	1.00 A
Dropout Voltage	0.5 V
Body Size	6.50 mm x 3.50 mm

Table 2: LM2940IMP-5.0/NOPB specifics.

The maximum output current needed would be

$$I_{\max} = I_{\mu C} + I_{M2} + I_{\text{USB}} = 9\,\text{mA} + 330\,\text{mA} + 15\,\text{mA} = 354\,\text{mA} < 1.00\,\text{A}$$

Being the maximum dropout voltage

$$V_{\text{DOmax}} = V_{\text{in}} - V_{\text{out}} = 7.4\,\text{V} - 5\,\text{V} = 2.4\,\text{V}$$

Grater than the dropout voltage of the chosen component (0.5 V) every requirement results satisfied.

BD60HC5MEFJ-M Low Dropout Regulator

Lastly for the second motor the BD60HC5MEFJ-M LDO voltage regulator by Rohm Semicunductor [2] was chosen accessible in a HTSOP-J8 type package (Figure 3). The following Table 3 is filled up with all utile the informations valid for a temperature range of -40 to +105 °C.

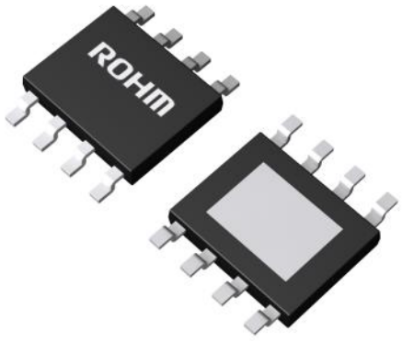


Figure 3: BD60HC5MEFJ

Item	Value
Input Voltage	from 4.5 V to 8 V
Output Voltage	6 V
Output current	1.5 A
Dropout Voltage	0.9 V
Body Size	6.00 mm x 4.90 mm

Table 3: BD60HC5MEFJ-M specifics.

The component dropout (0.9 V) results to be less than its max value ($7.4V - 6.0V = 1.4V$), equally can be said on its current ($1.00A < 1.50A$).

References

[1] TEXAS INSTRUMENTS. *LM2940 Low Dropout Regulator*. Datasheet. 2000. URL: https://www.ti.com/lit/ds/symlink/lm2940c.pdf.

[2] ROHM SEMICONDUCTOR. *Variable Output LDO Regulator BD60HC5MEFJ-M*. Datasheet. 2012. URL: https://xonstorage.blob.core.windows.net/pdf/rohm_bd70hc5vefjme2_apr22_xonlink.pdf.

Battery Attachment

GRUPPO 10

Battery

The battery represents the core power source of the spoon and has to furnish electrical voltage and current to all the integrated devices. Battery-devices paris take place through the employment of voltage regulators (for further informations see *Voltage Regulators Attachment*).

Criteria of Selection

Battery choice is built on technical requirements, dimensional constraints and design limitations. First of all it is essential that the battery can furnish each component the required electrical tension and that the output electrical current of the battery is greater than the sum of the input currents of the devices. For an appropriate electrical managment, the output current from the battery has to be at least 20% greater than the sum of the input currents of the devices powered (all the electrical specifications needed are included in Table 1 of *Voltage Regulators Attachment*). Therfore:

$$0.8 \cdot I_{PS} > 1000\,\text{mA} + 330\,\text{mA} + 15\,\text{mA} + 9\,\text{mA} = 1354\,\text{mA}$$

and so I_{PS} should be at least 1693 mA. Moreover battery dimenesions and mass must be as reduced as possible to permit the patient a more confortable and enjoyable meal.

Tenergy 18650

The selected batteries, Tenergy 18650, are rechargeable lithium cells, with a nominal capacity of 2200 mAh and a nominal voltage of 3.7 V each. To satisfy the project's requirement and correclty power all the components a configuration of two Tenergy 18650 connected in series was adopted, achieving a total voltage of $7.4V$. This choice was preferred over a single 7.4V battery for the following reasons:

- **Dimensional Constraints:** The 7.4V batteries available on the market generally have larger dimensions compared to 3.7V cells. Their height and diameter were incompatible with the space available in the mechanical CAD design of the project.

- **Optimized Space Usage:** Using two 3.7V batteries allowed arranging them in a way that minimizes space, keeping the configuration compatible with design constraints.
- **Flexibility in Placement:** The adoption of two separate batteries offers greater flexibility in their internal placement, facilitating integration into the overall device design.

Table 1 contains all the main technical specifications of the component.

Parameter	Value
Nominal capacity	2200 mAh (CA)
Minimum capacity	2100 mAh
Nominal voltage	3.7V
Internal impedance	$\leq 80m\Omega$ (with PTC)
Maximum charge voltage	$4.20 \pm 0.02V$
Standard charge current	0.5CA
Standard discharge current	0.5CA
Cycle life	≥ 300
Dimensions (Diameter $\times$ Height)	18.3 mm $\times$ 65.0 mm
Weight	45 g
Storage Temperature	0 - 35 °C

Table 1: Tenergy 18650 Battery Charateristics.

In the following subsections discharging time have been evaluted for a single battery but an equal dissertation leading to the same results can be done for the series of two identical batteries.

Discharging Time

Considering the battery nominal capacity and the maximum current needed to power the entire PCB, the discharging time of the battery, evaluating the quiescent current of the regulators negligible (few mA on an Ampere-ordered current), results:

$$t_{\mathrm{DIS}} = \frac{\mathrm{CA}}{I_{\mathrm{PCB}}} = \frac{2200\,\mathrm{mAh}}{1354\,\mathrm{mA}} \approx 1.62\,\mathrm{h}$$

Ergo a complete discharge happens in 1 hour and 37 minutes if the components' consumption is maximum. The discharging time allows the patient to have a complete meal which is esteemed to last around the hour.

Safety Performance Tests

A few safety performances tests, as in datasheet [1], were conducted and demonstrated the security of the product and the impossibility of fire and explosion even when submitted to extreme conditions.

References

[1] Tenergy Corporation. *18650 2200mAh Li-Ion Cell*. Datasheet. 2009. URL: https://www.jameco.com/Jameco/Products/ProdDS/2144243.pdf?srsltid=AfmBOoppo_Gw5frd9OBpqYeCdOfp2AolsmjnGagWIGXQPhE0VzTd1Djb.

Microcontroller Attachment

GRUPPO 10

ATmega328P-MUR

The ATmega328P-MUR (Figure 1) was chosen for the project due to its features, which make it particularly suitable for embedded devices. Having a very high maximun operating frequency (up to 20 MHz considering an external crystal) the microcontroller can execute stabilization algorithms, such as a complementary filter, in real-time without compromising performance. Communication with the IMU sensor is handled via the I2C interface, natively supported, ensuring smooth and reliable transmission of acceleration and angular velocity data. Additionally, the operational frequency of the I2C interface for both the microcontroller and the sensor is matched and sufficient to ensure accurate and complete data transmission. In fact, consider that for each sample, the sensor sends 2 bytes for each axis of the gyroscope and accelerometer. Additionally, since the IMU includes a temperature sensor, this data is also transmitted through another 2 bytes. So:

$$N_{Bfs} = 14 \,\text{bytes for sample}$$

Each byte is transmitted in 10 bits due to the 1 start bit and 1 stop bit. It follows that:

$$n_{bfs} = 14 \,\frac{\text{byte}}{\text{sample}} \cdot 10 \,\frac{\text{bit}}{\text{byte}} = 140 \,\frac{\text{bit}}{\text{sample}}$$

With a sampling frequency of 50 Hz, it results that:

$$Baud\,rate = 140 \,\frac{\text{bit}}{\text{sample}} \cdot 50 \,\frac{\text{sample}}{\text{s}} = 7000 \,\frac{\text{bit}}{\text{s}}$$

This is the minimum baud rate required for adequate transmission. The baud rate selected for data transmission between the sensor and the microcontroller is 9600.

The microcontroller offers low power consumption, which is essential for portable devices: it consumes approximately less then 1 mA/MHz in active mode, while in sleep mode, the consumption drops to just a few microamps, extending battery life. With six dedicated PWM channels, the ATmega328P-MUR allows precise and independent control of both motors used in the project, with an 8-bit resolution. The microcontroller also includes the necessary timers

for the device's operation. In particular, Timer 0 is used to calculate the transmission times of signals, while Timer 2 generates the PWM signals required for motor control [1]. Finally, the microcontroller is an economical and widely available solution, ideal for projects requiring cost efficiency. It is designed for surface-mount technology (SMD), making it easier to integrate into compact and optimized PCBs for portable devices.

Specification	**Details**
Architecture	AVR RISC 8-bit
Operating Frequency	Up to 20 MHz
Supply Voltage	1.8V - 5.5V
GPIO	23 total pins
Interfaces	I2C, SPI, UART
Timers	3 timers (timer0, timer1 and timer2)
PWM Channels	6 channels
Active Configuration Current at 5.0 V	9 mA
Idle Current at 5.0 V	2.5 mA
I2C Operating Frequency	100 kHz

Table 1: Specifications of the AVR Microcontroller



Figure 1: ATmega328P-MUR Microcontroller

References

[1] MICROCHIP. *megaAVR Datasheet*. Datasheet. 2020. URL: https://ww1.microchip.com/downloads/en/DeviceDoc/ATmega48A-PA-88A-PA-168A-PA-328-P-DS-DS40002061B.pdf.

Sensing Attachment

GRUPPO 10

Needed Sensing Charateristics

The previous data extraction lead to disponibility of the parameters shown in Table 1 in *Data Analysis Attachment*.

The maximum frequency registered has been on the z-axis of the accelerometer and results to be $f_{zmax} \sim 23$ Hz (considering its mean and SD) therefore for Nyquist-Shannon theorem a sampling frequency $f_s > 46$ Hz is required. The full-scale range of the selected sensor must also contain the minimum and maximum values registered for both the accelerometer and the gyroscope (available in Table 1 of *Data Analysis Attachment*). Other desired charateristics for the sensing module are the accuracy, sensitivity, durability, low power consumption and cost.

MPU-6050

The sensor selected is the IMU MPU-6050 [1] by IvenSense. This MotionTracking device that combines a 3-axis gyroscope and a 3-axis accelerometer encloses all the attributes cited in the previous section. All these aspects are summarized in Table 1 and refer to exercise temperatures between -40 and +85 °C.

Item	**Value**
Gyroscope Full-Scale Range	from ±250 to ±2000 °/s
Accelerometer Full-Scale Range	from ±2 to ±16 g
Active Configuration Current	3.9 mA
Idle Current	5 µA
Gyroscope Output Data Rate	from 4 to 8000 Hz
Accelerometer Output Data Rate	from 4 to 1000 Hz
I^2C Operating Frequency	100 kHz
Dimensions	4 mm x 4 mm x 0.9 mm

Table 1: MPU-6050 Charateristics.

The lowest full-scale ranges (±250 °/s for the gyroscope and ±2 g for the accelerometer) are ample for the application. By configuring the MPU-6050 with the lowest full-scale ranges, the sensor is

optimized for detecting the small, subtle movements characteristic of hand tremors, thereby enhancing the effectiveness of the tremor compensation system. As evident in Table 1 the programmable Gyroscope and Accelerometer Output Data Rates are fair to reach and even the communication Operating Frequency with the microcontroller is fully satisfactory.

The MPU-6050 (Figure 1) is a highly robust and durable sensor, designed to withstand a wide range of environmental conditions and mechanical stresses. Its shock tolerance of 10,000 g ensures that it can endure significant impacts without compromising performance, making it ideal for applications that require reliability and longevity. Additionally, the MPU-6050 features an integrated Digital Motion Processor (DMP), which offloads complex motion processing tasks from the host processor, enabling real-time motion tracking with minimal latency. Nevertheless its power consumption remains low in active mode when current absorption is just 3.9 mA.

This sensor is also remarkably compact, with a small form of $4 \times 4 \times 0.9$ mm, making it suitable for integration into space-constrained devices like a spoon. Despite its advanced features, the MPU-6050 is cost-effective, offering a high-performance solution at an affordable price point. Its extensive use in literature and various applications underscores its reliability and versatility.

Moreover, the MPU-6050 is designed for easy integration, supporting I2C interfaces, which simplifies the connection to a wide range of host systems. This ease of integration, combined with its comprehensive feature set, makes the MPU-6050 a popular choice for developers and engineers seeking a reliable, efficient, and cost-effective motion-sensing solution.

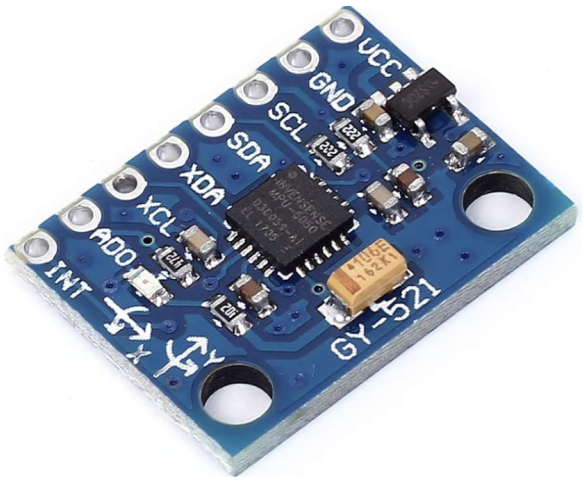


Figure 1: MPU6050 Sensor.

References

[1] IvenSense. *IMU MPU6050*. Datasheet. 2013. URL: https://invensense.tdk.com/wp-content/uploads/2015/02/MPU-6000-Datasheet1.pdf.